Phishing Detection & Awareness Report

An analysis of three email samples with red-flag identification and prevention guidance

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TRAINING REPORT • EMAIL THREAT AWARENESS

EXECUTIVE SUMMARY

This report analyses three email samples to demonstrate practical phishing detection techniques. Each sample is examined for sender spoofing, authentication failures, malicious links, urgency tactics, and social engineering indicators.

Results

1 email	**PHISHING — High risk**
1 email	**SUSPICIOUS — Medium risk**
1 email	**SAFE — No risk (sample classification)**

For each sample, this report provides:

- The identified red flags or green flags
- Business impact of the identified risk
- Clear prevention and awareness guidelines

The goal is to help employees recognise phishing attempts before they cause harm.

SCOPE & METHODOLOGY

Target Samples: Three email samples provided for training purposes

Analysis Date: 10 October 2026

Tools and Techniques Used

- Manual email header inspection
- Sender domain verification
- Link inspection (hover preview)
- Content analysis (urgency, greetings, tone)
- Social engineering indicator identification

Analysis Steps

1. Examine sender and reply-to addresses
2. Inspect the sending domain for spoofing or lookalikes
3. Identify urgency, threats, or pressure tactics
4. Examine any links or attachments mentioned
5. Check for generic vs personalised greetings
6. Classify the email as Safe / Suspicious / Phishing

Scope note: The samples are fictional training emails. Raw headers, live links, attachments, and authentication results were not provided; therefore SPF, DKIM, DMARC, and live link reputation were not independently verified.

FINDING 1 — PHISHING EMAIL

Sample 1: “Urgent: Email Account Will Be Suspended”

Classification: PHISHING **Risk Level: HIGH**

Sender: security-alert@example.invalid (Claimed: IT Support / IT Security Team)

Key Red Flags

- Sender domain uses “.invalid” TLD — reserved for testing and cannot belong to a real organisation.
- Extreme urgency: “within the next 30 minutes”.
- Threatens serious consequences: “permanent suspension” and “loss of access to your mailbox and stored files”.
- Generic greeting: “Dear Employee” instead of a real name.
- Impersonation of IT Support / IT Security Team.
- Vague verification link: “[Training link — do not use a real URL]”.
- Prohibits replies: “Please do not reply to this email”, discouraging verification.

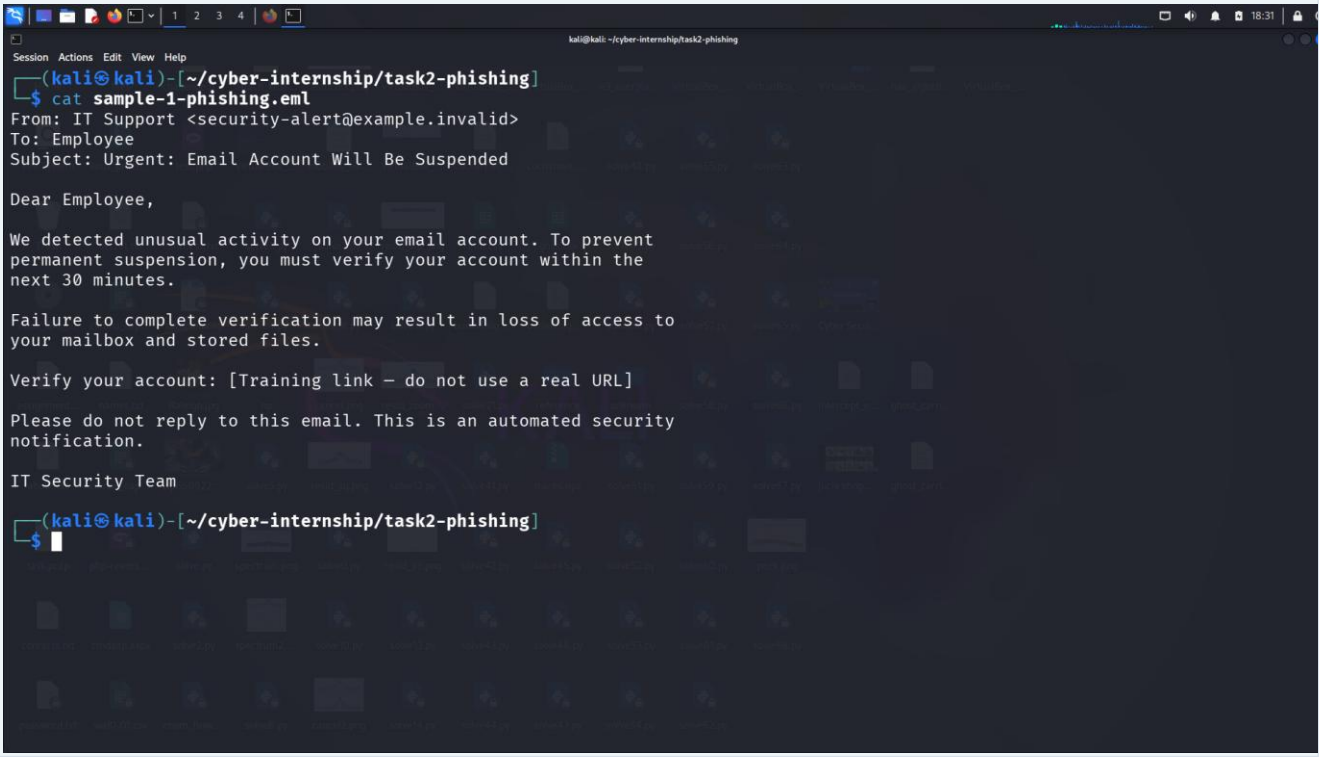
Business Impact

- If credentials are entered after clicking a malicious link, attackers could access the corporate mailbox.
- Attackers could read or exfiltrate stored files.
- A compromised account could be used to pivot into the wider corporate network, depending on access and controls.
- Attackers could send further phishing emails from the compromised account.

Prevention Guidelines

- Verify the sender’s actual domain, not just the display name.
- Do not click unsolicited links or enter credentials through email links.
- Report the email to IT Security through a known, trusted channel.
- Use an independently accessed official site or application to check account status.

Evidence



FINDING 2 — SAFE EMAIL

Sample 2: “Meeting Scheduled: Security Team Review”

Classification: SAFE SAMPLE **Risk Level: NONE (illustrative)**

Sender: security-team@company.example (Presented as an internal Security Operations Team)

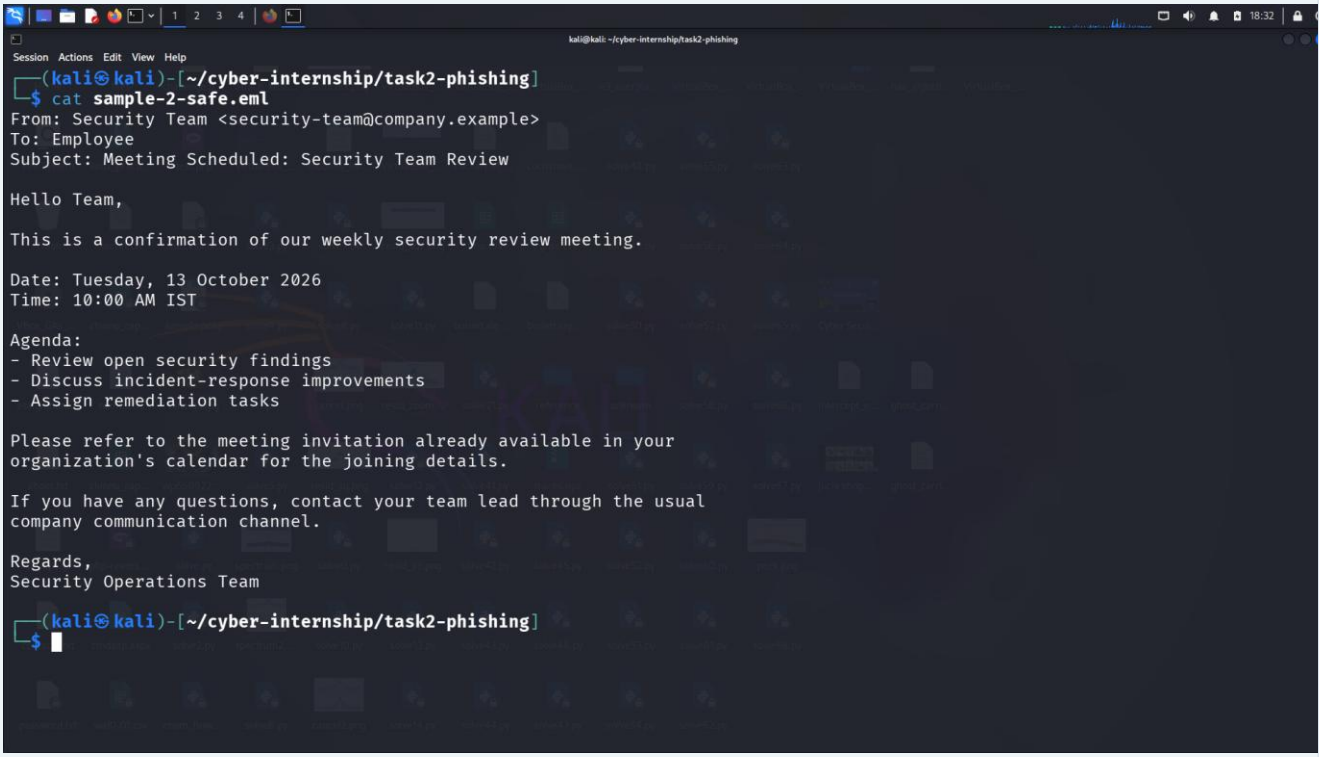
Key Green Flags

- Sender domain uses “.example”, a reserved domain for documentation and examples.
- Subject is routine: a weekly team meeting confirmation.
- No urgency or threatening language.
- Meeting is scheduled for 13 October 2026, without pressure to act immediately.
- Greeting “Hello Team” is appropriate for a team-wide notice.
- References an existing calendar invitation rather than introducing a new link.
- Provides a verification path through the usual company communication channel.
- No suspicious attachments or credential-harvesting links are included in the sample.

Conclusion

This email shows the characteristics legitimate internal notification because it contains routine meeting information and directs the recipient to an existing calendar invitation or known internal channel. The reserved example domain is not proof of a real sender. In a real investigation, verify the sender, headers, and context before declaring an email safe.

Evidence



FINDING 3 — SUSPICIOUS EMAIL

Sample 3: “Action Required: Pending Invoice Approval”

Classification: SUSPICIOUS **Risk Level: MEDIUM**

Sender: accounts-review@example.invalid (Claimed: Accounts Department)

Warning Signs

- Unexpected financial request for INR 48,750.
- Creates urgency: “payment is scheduled for processing today”.
- Sender domain uses “.invalid”, reserved for testing, not a real business domain.
- Generic greeting: “Hello” instead of a specific name.
- Mentions an attachment (“the attached invoice”), which could be malicious.
- Impersonates an internal Accounts Department.
- Pressure to act quickly to “avoid a delay”.

Some Positive Signs

- Includes a suggestion to verify through the organisation’s verified finance contact details.
- Does not include a direct credential-harvesting link.
- The wording is professionally consistent.

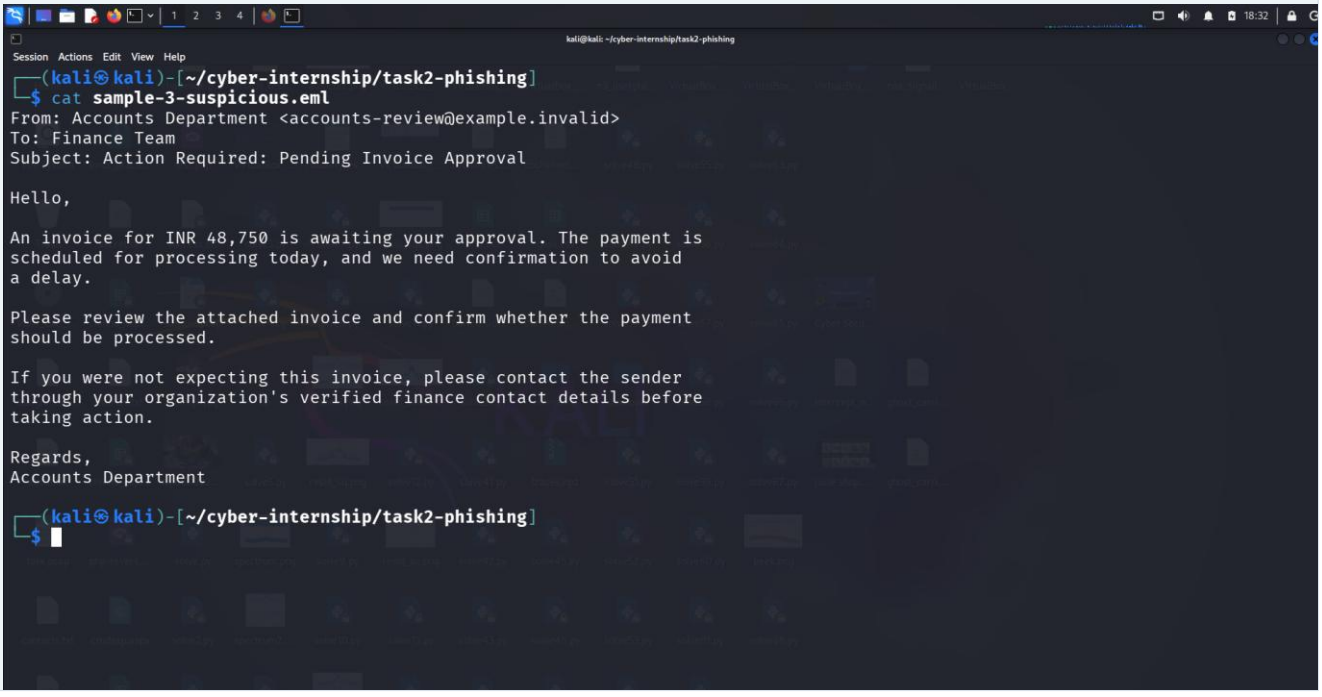
Why Suspicious, Not Confirmed Phishing

- The request could resemble a legitimate invoice workflow, but sender and request have not been independently verified.
- No phishing link is provided in the sample; it mentions only an attachment.
- The security-conscious verification wording is a positive signal, but does not establish legitimacy.

Recommended Actions

- Do NOT open the attachment without verification.
- Contact Finance through a known, trusted phone number — not by replying to the email.
- Report the email to IT Security for verification.
- Do not click links or reply until the message is verified.

Evidence



PHISHING AWARENESS GUIDELINES FOR USERS

1. Verify the Sender

Check the actual email address, not just the display name. Look for misspellings or unusual domains. A domain alone does not prove that a sender is legitimate.

2. Beware of Urgency

Phishing emails often demand immediate action — “within 30 minutes”, “today”, or “final warning”. Treat pressure as a reason to verify, not as proof by itself.

3. Examine the Greeting

“Dear Employee” or “Dear Customer” can be a warning sign in context, but generic greetings are not proof of phishing.

4. Inspect Links Before Clicking

Hover over links to preview the destination where possible. Visible text can differ from the actual URL. HTTPS alone does not guarantee a site is trustworthy.

5. Treat Unexpected Attachments with Caution

Do not open unexpected invoices, receipts, or documents without verifying the sender through a trusted channel.

6. Never Enter Credentials via Email Links

Navigate to the official site yourself by typing the known URL or using a trusted bookmark.

7. Verify Financial Requests Independently

For payment or invoice approval, call the requester using a known number — do not rely only on contact details in the message.

8. Report Suspicious Emails

Forward the email to your IT/security team or use the “Report Phishing” feature in your email client. Follow your organisation’s reporting process.

CONCLUSION

This analysis demonstrated how to identify phishing emails through sender verification, content analysis, and social engineering awareness.

Key Takeaways

- ".invalid" and ".example" are reserved domains for testing and documentation; they are not evidence of real organisational senders.
- Urgency, threats, and generic greetings are useful indicators to examine, but no single indicator proves phishing.
- Not every suspicious email is confirmed phishing — some require verification through trusted channels before classification.
- Safe-looking emails should still be verified using sender details, headers, and expected business context.

Phishing remains a common initial attack vector. Consistent user awareness and independent verification habits are essential defences.